Project Exploration Assignment Guidelines

COMP 468/568 – Spring 2026

In this assignment, you will explore an existing open-source GitHub project related to Deep Learning Systems (e.g., training systems, inference engines, compilers, distributed ML, or system optimizations). The goal is to understand a real-world ML systems artifact, reproduce its results, and critically analyze its design and performance.

1. Project Signup

Each student must sign up for ONE project using the provided Google Sheet.
https://docs.google.com/spreadsheets/d/1A6mRKnstF_ZnnTlxThOGPJXnHrSnx1dldHg4RKAxhNk/edit?usp=sharing

You must:
- Select a GitHub repository related to deep learning systems
- Enter your name(s) and the project link in the Google Sheet
- Ensure no duplicate selections unless explicitly allowed

Once selected, you are responsible for exploring and evaluating that artifact.

2. Background and Artifact Description

Provide background information about the selected project:

- What problem does this system address?
- What type of artifact is it (e.g., training framework, compiler, runtime, optimizer)?
- What are the key ideas or system contributions?
- Why is this project relevant to deep learning system design?

Cite the GitHub repository and any associated papers or documentation.

3. Code Setup and Environment

Describe how you set up the project so others can reproduce your work.

Include:
- Hardware used (CPU/GPU type, memory)

- Software environment (OS, Python version, framework versions)
- Installation steps and dependencies
- Any issues encountered during setup and how you resolved them

4. Evaluation Platform

Clearly state where and how the project was evaluated:

- Local machine, shared server, cloud VM, or cluster
- Number of GPUs or nodes used
- Any platform-specific constraints or configurations

Explain why this platform is suitable for evaluating the artifact.

5. Implementation and Deployment

Explain how you ran, integrated, or modified the code.

Include:
- Which scripts or entry points were used
- Any configuration changes or code modifications you made
- How the system was executed (training, inference, benchmarking)
- Deployment details if applicable (e.g., containers, distributed launch)

6. Screenshots of Results

Include 2–4 screenshots demonstrating the artifact in action.

Examples:
- Execution logs or terminal output
- Performance metrics (runtime, throughput, memory)
- Profiler timelines or dashboards
- Training or inference progress

Each screenshot must be readable, captioned, and referenced in the report.

7. Results

Summarize the main results you obtained.

Include:
- Key performance metrics
- Comparison to any baseline provided by the project
- Whether your results match those reported by the authors

Use tables or figures if helpful.

8. Observations

Discuss what you observed from running and evaluating the system.

Examples:
- Performance trends or bottlenecks
- Resource utilization behavior
- Sensitivity to configuration choices

Focus on explaining why the system behaves the way it does.

9. Improvements and Extensions

Describe any improvements or optimizations you attempted or proposed.

You may include:
- Changes you implemented and their impact
- Optimizations you attempted but did not finish
- Ideas for future improvements based on your observations

Clearly state what was implemented versus what is proposed.

10. Code Submission (Required)

You must submit:
- A link to the original GitHub project
- Your modified code or scripts (if any)
- Clear instructions to reproduce your results

All code must run without errors. Missing or non-runnable code will result in reduced credit.

Note: Failure to sign up in the Google Sheet, missing screenshots, or missing code submission will be considered an incomplete assignment.